

YZV202E Optimization for Data Science

Final Project Report

Multi-Criteria Location Optimization for Data Center Placement in Türkiye

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Abstract—Choosing where to build a data center involves balancing many competing factors at once: seismic risk, cooling costs, land availability, workforce size, and renewable energy access. In this project we tackle this problem for Türkiye’s 81 provinces by formulating it as a Binary Integer Program (BIP). Each province receives a composite suitability score from five normalized criteria, and the solver picks the best set of cities that maximizes total suitability while respecting constraints on geographic separation, seismic safety, and proximity to the southern border. We collected real data from eight public sources—AFAD, TÜİK, MGM, NASA POWER, TÜREB, SRTM, KGM, and OpenStreetMap—implemented the model in Python with PuLP/CBC, and deployed an interactive Streamlit dashboard where users can adjust weights and constraints in real time. Under default settings with equal criterion weights and $K = 5$, the model selects Ankara, Kayseri, Erzurum, Yozgat, and Aksaray. We also demonstrate how the solution shifts meaningfully under seismic-priority, cooling-priority, and renewable-priority weight profiles, confirming the model’s responsiveness to user preferences.

I. INTRODUCTION

Cloud computing and artificial intelligence workloads have been growing steadily over the past decade, and with that growth comes increasing demand for physical data center infrastructure in new regions. Türkiye sits at the intersection of Europe and Asia, has a young and growing population, and has been investing heavily in its digital backbone. These features make it an attractive candidate for global technology companies looking to expand.

However, picking the right city is not straightforward. A good data center location needs flat terrain for construction, low seismic risk to protect expensive hardware, a large enough local workforce, affordable and ideally renewable energy, and a cold climate to keep cooling costs down. The trouble is that these criteria often conflict. İstanbul has the biggest labor pool in the country, but it sits right on the North Anatolian Fault and its summers are warm. Ardahan has the coldest climate in Türkiye—ideal for cooling—but almost nobody lives there.

The goal of this project is to handle these trade-offs systematically through optimization. We frame the data cen-

ter placement problem as a multi-criteria facility location task, formulated as a Binary Integer Program over Türkiye’s 81 provinces. The entire pipeline—from raw data collection to interactive visualization—is implemented in Python and deployed as a Streamlit web application where users can explore different scenarios by adjusting weights, constraints, and solver modes on the fly.

The remainder of this report is organized as follows. Section II defines the mathematical model. Section III describes the datasets and the normalization applied to each criterion. Section IV details the solving methodology, including the branch-and-bound algorithm and the interactive application. Section V presents results under multiple weight scenarios. Section VI discusses limitations, and Section VII concludes.

II. PROBLEM DEFINITION

Out of Türkiye’s 81 provinces, our goal is to identify the optimal cities for data center investment. Each province is represented by a single candidate point—its official city center coordinates retrieved from OpenStreetMap—providing a reproducible representative location per province.

A. Suitability Score

Each province i (indexed 1–81, following the Turkish license plate numbering) receives a composite suitability score:

$$x_i = w_1 F_i + w_2 P_i + w_3 L_i + w_4 E_i + w_5 C_i \quad (1)$$

where $F_i, P_i, L_i, E_i, C_i \in [0, 1]$ are normalized scores for land availability, seismic safety, labor density, renewable energy potential, and cooling efficiency, respectively. The weights satisfy $\sum_{j=1}^5 w_j = 1$ with $w_j \geq 0$. The computation of each criterion is detailed in Section III.

B. Decision Variables and Objective Function

Let $y_i \in \{0, 1\}$ be a binary decision variable for province i , where $y_i = 1$ if a data center is placed there. The application supports two selection modes.

Count Mode. The user specifies the number of provinces K to select:

$$\text{maximize } Z = \sum_{i=1}^{81} x_i y_i \quad \text{s.t.} \quad \sum_{i=1}^{81} y_i = K \quad (2)$$

Budget Mode. The model selects as many provinces as the budget allows:

$$\text{maximize } Z = \sum_{i=1}^{81} x_i y_i \quad \text{s.t.} \quad \sum_{i=1}^{81} c_i y_i \leq B \quad (3)$$

where c_i is the estimated cost for province i and B is the user-specified budget ceiling. In both modes, the objective and constraints are linear functions of binary variables, placing the problem in the Binary Integer Programming family.

C. Constraints

Both modes are subject to the following additional constraints:

(C1) Minimum distance. For disaster recovery, any two selected provinces must be at least D km apart in road network distance. We pre-compute the set of forbidden pairs from the KGM distance matrix:

$$\mathcal{F} = \{(i, j) \mid i < j, d_{ij} < D\} \quad (4)$$

and enforce:

$$y_i + y_j \leq 1, \quad \forall (i, j) \in \mathcal{F} \quad (5)$$

Pre-computing forbidden pairs is essential to preserving linearity. A naïve formulation involving distance directly (e.g., $d_{ij} \cdot y_i \cdot y_j \geq D$) would introduce quadratic terms. By pre-filtering, we convert this into a set of simple linear inequalities.

(C2) Seismic safety threshold. The total seismic safety score of selected provinces must exceed a minimum $R_{\min}$:

$$\sum_{i=1}^{81} P_i y_i \geq R_{\min} \quad (6)$$

(C3) National security (border exclusion). Türkiye's southern border with Syria and Iraq runs roughly along the 37th parallel [15]. Provinces whose center is closer than D_{safe} km to this reference latitude are excluded:

$$y_i = 0, \quad \forall i : R \cdot |\phi_i - \phi_{\text{border}}| < D_{\text{safe}} \quad (7)$$

where $R = 6371$ km, ϕ_i is province i 's latitude in radians, and $\phi_{\text{border}} = 37 \times \pi/180$. The 37th-parallel reference is a deliberate simplification: the actual border deviates by at most ~ 50 km along the entire southern frontier, with the largest deviations in the easternmost provinces (Hakkari, Şırnak). Since D_{safe} is user-adjustable via a slider, any conservative estimation can be compensated by increasing it. A polyline-based exact-distance computation against the full border geometry was considered but not adopted, as it would not change the eliminated set under realistic D_{safe} values (100–200 km) while adding preprocessing complexity.

(C4) Binary domain:

$$y_i \in \{0, 1\}, \quad \forall i \in \{1, \dots, 81\} \quad (8)$$

III. DATASET

All data was collected from publicly available sources and processed into five normalized criterion scores plus a cost estimate for each of the 81 provinces. The final processed dataset contains 81 rows and 10 columns (province ID, name, latitude, longitude, F , P , L , E , C , cost). Below we describe each criterion, its source, and the exact normalization.

A. Land Availability (F)

Source: NASA SRTM 3-arc-second (≈ 90 m) digital elevation model, with province boundaries from GADM v4.1.

Processing: Türkiye's terrain varies significantly in elevation, so not all provinces offer sufficient flat land for large-scale infrastructure. For each province, we computed pixel-wise terrain slope from the elevation raster. Let A_i be the total area (km^2) of province i with slope below a buildable threshold θ :

$$A_i = |\{p \in \text{province}_i \mid s_p < \theta\}| \cdot r^2 \quad (9)$$

where s_p is the slope at pixel p and $r \approx 0.09$ km is the SRTM grid resolution. The default threshold is $\theta = 5^\circ$. Standard min-max normalization is applied:

$$F_i = \frac{A_i - A_{\min}}{A_{\max} - A_{\min}} \quad (10)$$

Konya dominates with $F = 1.0$ (nearly 30 000 km^2 of flat terrain on the Konya plain), while Rize scores $F = 0.0$ —its entire area is steep mountain terrain along the Black Sea coast.

B. Seismic Safety (P)

Source: AFAD (Disaster and Emergency Management Presidency) Turkish Seismic Hazard Map—Peak Ground Acceleration (PGA) values at fixed-spacing grid points for the “10% exceedance in 50 years” scenario, which is the standard approach in seismic hazard assessment [1].

Processing: Each PGA grid point was spatially matched to its enclosing province, and the mean PGA was computed per province. Since PGA values are well-established in the seismic hazard literature to follow an approximately log-normal distribution [12], [13], logarithmic normalization yields more accurate scaling. We invert so that lower PGA (safer) maps to a higher score:

$$P_i = 1 - \frac{\ln(\text{PGA}_i) - \ln(\text{PGA}_{\min})}{\ln(\text{PGA}_{\max}) - \ln(\text{PGA}_{\min})} \quad (11)$$

where $\text{PGA}_{\max}$ and $\text{PGA}_{\min}$ are the maximum and minimum mean-PGA values observed across all 81 provinces. The PGA values range from 0.09 g (Nevşehir, $P = 1.0$) to 0.60 g (Bingöl, $P = 0.0$).

C. Labor Density (L)

Source: TÜİK (Turkish Statistical Institute)—provincial employment rates (η_i) and 15+ age population (N_i).

Processing: The raw labor force is the product of working-age population and employment rate:

$$L_{\text{raw},i} = N_i \cdot \eta_i \quad (12)$$

Since labor suitability does not scale linearly with raw population size—the difference between 50 000 and 100 000 workers matters more than between 5 million and 5.05 million—we apply a log transform before normalization:

$$L_{\text{adj},i} = \ln(L_{\text{raw},i}), \quad L_i = \frac{L_{\text{adj},i} - L_{\text{adj},\min}}{L_{\text{adj},\max} - L_{\text{adj},\min}} \quad (13)$$

İstanbul leads with $L = 1.0$ (~ 7.6 million employed), followed by Ankara ($L = 0.80$) and İzmir ($L = 0.75$). Bayburt, the smallest province, scores $L = 0.0$. Across all weight presets, the labor criterion is intentionally assigned a relatively low weight, reflecting the fact that skilled IT personnel can relocate and labor availability should not dominate the placement decision.

D. Renewable Energy Potential (E)

Sources: NASA POWER API for annual average Global Horizontal Irradiance (GHI, W/m^2); TÜREB (Turkish Wind Energy Association) July 2022 Statistical Report for installed wind capacity (MW) per province.

Processing: The provincial wind capacity distribution is heavily skewed—a few provinces like İzmir (1 907 MW) and Balıkesir (1 526 MW) host the bulk of installed capacity while many provinces have zero. To handle this, we apply a logarithmic adjustment with a +1 shift:

$$W_{\text{adj},i} = \ln(W_{\text{raw},i} + 1) \quad (14)$$

Solar GHI and adjusted wind are each min-max normalized to obtain $\tilde{S}_i$ and $\tilde{W}_i$, then combined:

$$E_i = \alpha \tilde{S}_i + \beta \tilde{W}_i, \quad \alpha + \beta = 1 \quad (15)$$

The default internal weights are $\alpha = 0.6$ (solar) and $\beta = 0.4$ (wind), adjustable via the application interface. Solar potential is uniformly available across all provinces while wind is concentrated in the Aegean and Marmara corridors, justifying the slight solar emphasis. Using installed capacity rather than raw wind speed is a deliberate choice: high installed capacity simultaneously signals favorable wind conditions, existing grid integration, and proven investor interest—all directly relevant to data center siting. İzmir scores highest ($E = 0.89$); Artvin scores $E = 0.0$.

E. Cooling Efficiency (C)

Source: MGM (Turkish State Meteorological Service)—long-term average annual temperatures per province. Because no structured data file was publicly available, these values were obtained via web scraping.

Processing: Lower temperature means less cooling energy, so we invert and normalize:

$$C_i = 1 - \frac{T_i - T_{\min}}{T_{\max} - T_{\min}} \quad (16)$$

Ardahan scores $C = 1.0$ (annual average 4.3°C at $\sim 1\,800$ m elevation); Mersin on the Mediterranean scores $C = 0.0$ (20.2°C). The full range is 4.3 – 20.2°C .

F. Cost Estimation (c_i)

Sources: TÜİK provincial GDP (NUTS-3, 2024); EPDK (Energy Market Regulatory Authority) industrial electricity tariff (April 2026).

Processing: Each province’s estimated data center cost combines a constant base, regional personnel expenses, and national energy costs:

$$c_i = C_{\text{base}} + P_{\text{exp}} \cdot I_i + E_{\text{annual}} \cdot U_{\text{elec}} \quad (17)$$

where $C_{\text{base}} = 50\text{M TL}$ is a fixed construction baseline, $P_{\text{exp}} = 100$ is the expected headcount, I_i is the local average income derived from provincial GDP, and $E_{\text{annual}} \cdot U_{\text{elec}}$ captures annual electricity cost at the EPDK industrial tariff of ≈ 3.98 TL/kWh. Costs range from 90M TL (Bayburt) to 5.1B TL (İstanbul).

G. Distance Matrix and Coordinates

Inter-provincial road distances were obtained from KGM (General Directorate of Highways) as an 81×81 symmetric matrix. Province center coordinates come from OpenStreetMap via the `osmnx` Python library. Province boundary polygons for choropleth visualization come from GADM v4.1, mapped through NUTS-3 codes.

H. Summary Statistics

Table I shows descriptive statistics across all 81 provinces. A notable pattern is the negative correlation between labor density and cooling efficiency ($\rho = -0.45$): large cities with big labor pools tend to be warmer, while cold provinces tend to have small populations. This tension is central to the optimization.

TABLE I
DESCRIPTIVE STATISTICS OF CRITERION SCORES ($n = 81$).

Criterion	Mean	Std	Min	Max
F (Land)	0.123	0.139	0.000	1.000
P (Seismic)	0.432	0.221	0.000	1.000
L (Labor)	0.353	0.184	0.000	1.000
E (Energy)	0.409	0.171	0.000	0.887
C (Cooling)	0.417	0.206	0.000	1.000

IV. METHODOLOGY

A. Why Binary Integer Programming?

Selecting the right optimization method requires matching the method to the problem structure. Our problem has several clear features: the decision is purely combinatorial (choosing a subset of cities, not tuning continuous parameters), the objective function is a weighted sum of fixed scores (linear in the decision variables), and all constraints can be expressed as linear equalities or inequalities over binary variables. This places the problem naturally in the Binary Integer Programming (BIP) framework.

We considered genetic algorithms (GA) during the planning phase. A GA could handle the discrete selection and would allow for non-linear extensions in the future. However, since

both the objective and all constraints are fully linear and the search space contains only 81 binary variables, an exact BIP solver finds the provably global optimum in under a second. A heuristic like GA would offer no advantage here—it would trade guaranteed optimality for stochastic approximation without any speed benefit for this problem size. If the problem were scaled up to, say, thousands of candidate locations or if non-linear constraints were introduced, a GA or other metaheuristic would become a more reasonable choice.

B. Branch and Bound

The BIP is solved using the Branch and Bound (B&B) algorithm, which is the standard exact method for integer programming. The key idea is to systematically explore the solution space by solving a sequence of Linear Programming (LP) relaxations. The algorithm proceeds as follows:

- 1) **LP Relaxation.** Replace the binary constraint $y_i \in \{0, 1\}$ with $0 \leq y_i \leq 1$. Solve this continuous LP to get an upper bound on the optimal objective value.
- 2) **Branching.** If the LP solution has fractional values (e.g., $y_3 = 0.7$), pick one such variable and create two subproblems: one where $y_3 = 0$ and one where $y_3 = 1$.
- 3) **Bounding.** Solve the LP relaxation of each subproblem. If the upper bound of a subproblem is worse than the best known integer solution, prune that branch—no integer solution in that subproblem can improve on the incumbent.
- 4) **Termination.** Continue branching and pruning until all subproblems are either solved or pruned. The best integer-feasible solution found is guaranteed to be globally optimal.

Although B&B has worst-case exponential complexity in the number of binary variables (2^{81} nodes in the worst case for our problem), the LP relaxation bounds are typically tight enough to prune the vast majority of branches. In practice, the CBC solver handles our 81-variable instances in well under one second, even with hundreds of distance-pair constraints active.

C. Solver Implementation

We used PuLP [10], an open-source Python LP modeling library, with the CBC (Coin-or Branch and Cut) solver [11] as the backend. CBC extends basic B&B with cutting planes—additional linear inequalities that tighten the LP relaxation without eliminating any integer-feasible solutions. This combination of branching and cutting makes CBC particularly effective for binary programs like ours.

The implementation follows a five-step pipeline:

- 1) **Data loading.** The pre-processed scores file (81 rows, 10 columns) is loaded into a Pandas DataFrame. Solar and wind sub-scores are loaded separately so the α/β blend can be recomputed at runtime.
- 2) **Score computation.** The suitability score x_i is computed using user-specified weights. Weights are entered as raw integers through sliders and auto-normalized to sum to 1.

- 3) **Preprocessing.** The KGM distance matrix is filtered to produce the forbidden-pair set $\mathcal{F}$. Border-excluded provinces are identified. These are computed once per parameter configuration.
- 4) **Model construction and solving.** Binary variables y_i , the objective $\sum x_i y_i$, and all constraints are added to the PuLP model. CBC solves the BIP, typically in under one second.
- 5) **Visualization.** Selected provinces are highlighted on a choropleth map of Türkiye rendered using GeoPandas and Matplotlib inside the Streamlit interface (Fig. 1).

D. Weight Determination

Determining appropriate criterion weights is a critical modeling choice. In our proposal, we described the Analytic Hierarchy Process (AHP) as a principled approach for deriving weights from pairwise comparisons. In the final implementation, we opted for a more flexible approach: rather than locking in a single AHP-derived weight vector, we provide both preset weight templates and fully adjustable sliders, enabling the user to test multiple scenarios.

The preset templates are: *Equal Weights* (all criteria at 0.20), *Seismic Priority* ($w_P = 0.60$, others at 0.10), *Cooling Priority* ($w_C = 0.60$, others at 0.10), *Renewable Priority* ($w_E = 0.60$, others at 0.10), and *Land Priority* ($w_F = 0.60$, others at 0.10). Users can also set arbitrary weights through the sliders (Fig. 2). This design was motivated by the instructor’s feedback that the interface should support easy comparison across weight scenarios.

E. Interactive Application

The Streamlit dashboard (Fig. 1) exposes all tunable parameters through sidebar controls (Fig. 2): solar/wind blend ratio (α/β), land slope threshold θ , five criterion weight sliders with preset templates, selection mode toggle (count K or budget B), and constraint parameters D , $R_{\min}$, and D_{safe} . When the user clicks “Solve!,” the BIP is reconstructed and solved in real time, and the choropleth map updates to display the optimal provinces colored by suitability score. A result table beneath the map lists each selected province with its individual criterion scores and estimated cost.

V. RESULTS

A. Baseline Scenario

Under default settings—equal weights ($w_j = 0.2$ for all j), $K = 5$, $D = 100$ km, $R_{\min} = 1.0$, $D_{\text{safe}} = 100$ km—the solver selects the provinces in Table II. All five are inland: four on the Central Anatolian plateau (Ankara, Kayseri, Yozgat, Aksaray) and one in the east (Erzurum). Ankara brings the strongest labor pool ($L = 0.80$), Aksaray the best seismic safety ($P = 0.86$), and Erzurum the highest cooling score ($C = 0.93$).

Konya, which has the highest individual suitability ($x_i = 0.692$) under equal weights, is *not* selected because its latitude (37.87°N) places it inside the border exclusion zone.

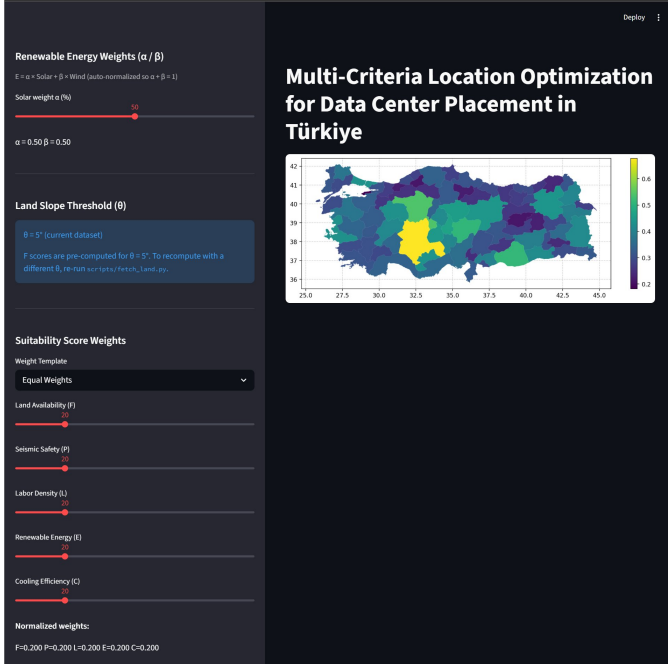


Fig. 1. Streamlit dashboard showing the choropleth map with equal-weight suitability scores, sidebar controls for energy blend and criterion weights.

TABLE II
BASELINE: $K = 5$, EQUAL WEIGHTS (0.20 EACH).

ID	Province	x_i	F	P	L	E	C
6	Ankara	0.555	0.42	0.66	0.80	0.42	0.48
38	Kayseri	0.522	0.28	0.68	0.51	0.56	0.58
66	Yozgat	0.471	0.23	0.74	0.30	0.43	0.66
68	Aksaray	0.468	0.21	0.86	0.29	0.51	0.46
25	Erzurum	0.460	0.18	0.29	0.40	0.50	0.93
Total Z		2.476					

B. Weight Scenario Comparison

To evaluate the model’s sensitivity to user preferences, we ran four weight profiles with $K = 5$ and identical constraints. Table III summarizes the selected cities under each profile.

The results show clear and intuitive shifts:

Seismic Priority concentrates selections in central Anatolia, the safest seismic zone in Türkiye. Kırşehir ($P = 0.87$) enters the set despite its small population—the high seismic weight overrides labor concerns. The total Z is highest (3.131) because the seismic scores in this region are very high and contribute disproportionately to the weighted sum.

Cooling Priority shifts the solution dramatically eastward to the high-altitude plateau: Erzurum ($C = 0.93$), Kars ($C = 0.93$), and Ağrı ($C = 0.85$) all appear. These provinces have average annual temperatures between 4.3°C and 6.7°C . The trade-off is visible in their low labor and energy scores.

Renewable Priority is the only scenario that selects a coastal city: İzmir enters the solution with the highest energy score ($E = 0.89$) thanks to its 1 907 MW of installed wind capacity and strong solar irradiance. Balıkesir (1 526 MW wind) also

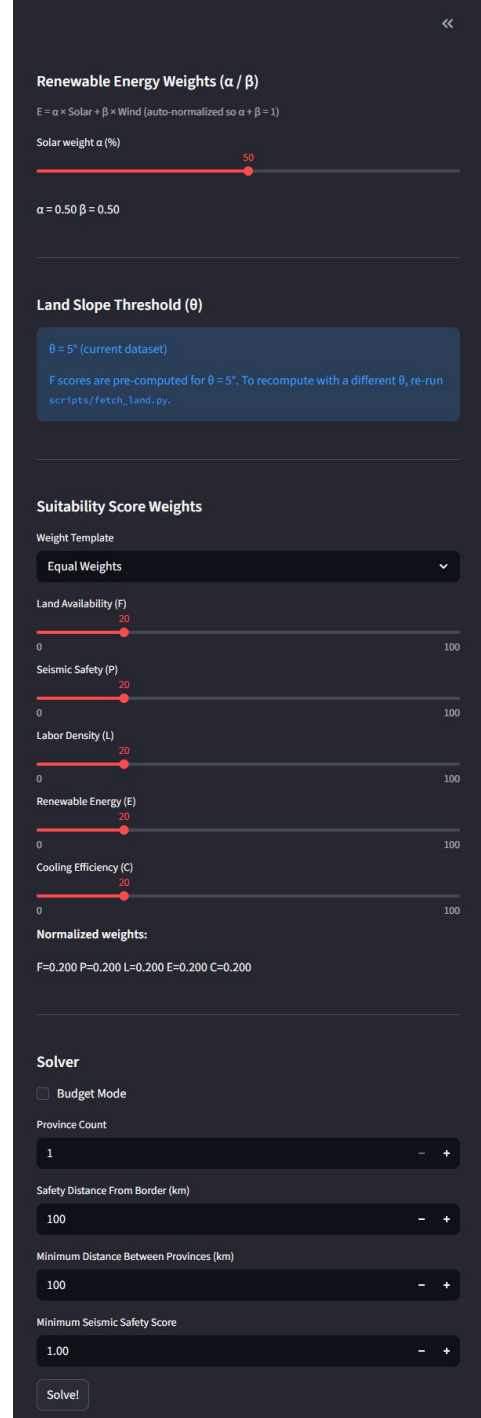


Fig. 2. Sidebar interface: weight sliders, weight template selector, solver mode toggle (count/budget), and constraint parameter inputs (K , D_{safe} , D , R_{min}).

appears. This scenario demonstrates how prioritizing renewables pulls the solution toward the Aegean coast, which is geographically very different from the baseline.

TABLE III
SELECTED PROVINCES UNDER FOUR WEIGHT SCENARIOS ($K = 5$, $D = 100$ km, $R_{\min} = 1.0$, $D_{\text{SAFE}} = 100$ km).

Scenario	City 1	City 2	City 3	City 4	City 5	Total Z
Equal (0.20 each)	Ankara	Kayseri	Yozgat	Aksaray	Erzurum	2.476
Seismic ($w_P = 0.60$)	Aksaray	Kırşehir	Ankara	Yozgat	Kayseri	3.131
Cooling ($w_C = 0.60$)	Erzurum	Kars	Ağrı	Yozgat	Sivas	3.123
Renewable ($w_E = 0.60$)	İzmir	Kayseri	Balıkesir	Van	Aksaray	2.680

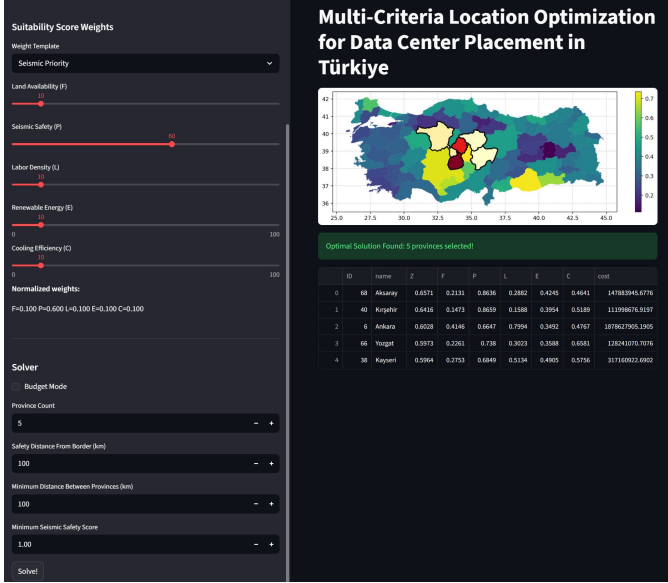


Fig. 3. Seismic Priority ($w_P = 0.60$, $K = 5$): Aksaray, Kırşehir, Ankara, Yozgat, and Kayseri are selected. High suitability (yellow) concentrates in central Anatolia where PGA values are lowest.

C. Effect of Constraints

The constraints play an important role in shaping solutions beyond what weights alone determine:

Distance constraint. At $D = 100$ km, 38 province pairs are forbidden. Increasing D to 200 km raises this to over 150 pairs, forcing the solver to spread selections geographically and sometimes sacrificing higher-scoring but clustered provinces. For instance, at $D = 200$ km, Ankara and Kırşehir cannot both be selected (road distance: 186 km).

Border exclusion. The default $D_{\text{safe}} = 100$ km eliminates 21 provinces. This removes strong multi-criteria performers like Konya ($x_i = 0.692$ under equal weights), Şanlıurfa ($x_i = 0.531$), and Antalya from the feasible set. If D_{safe} is relaxed to 50 km, Konya enters the baseline solution and pushes total Z from 2.476 to approximately 2.55.

Seismic threshold. The default $R_{\min} = 1.0$ is permissive—easily satisfiable when $K = 5$, since many provinces have $P > 0.20$. Raising $R_{\min}$ to 3.0 or higher begins to bind, forcing the solver to select at least some high- P provinces even if other criteria would prefer different cities.

VI. LIMITATIONS

We want to be transparent about the model’s limitations, several of which we identified during the proposal phase and

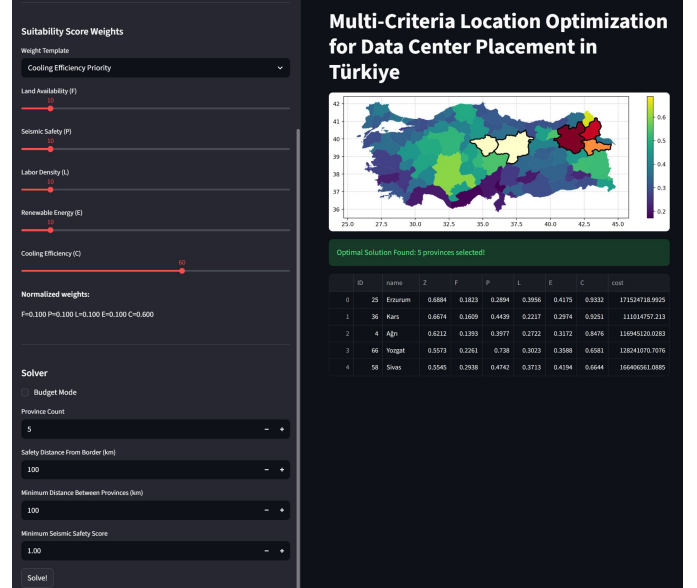


Fig. 4. Cooling Priority ($w_C = 0.60$, $K = 5$): Erzurum, Kars, Ağrı, Yozgat, and Sivas are selected. The solution shifts dramatically eastward to the high-altitude plateau where annual temperatures range from 4.3°C to 13.2°C.

deliberately chose not to address for scope reasons.

Province-level granularity. All scores are based on province averages computed from the official city center. A real data center would be sited at a specific parcel, and conditions (terrain, seismic risk, temperature) can vary considerably within a province, especially large ones. As a noted side effect, the model already prefers Central Anatolia for land availability, and the border constraint reinforces this by excluding southern provinces—potentially biasing the model toward a single region.

NEET and youth unemployment. Youth unemployment and Eurostat’s NEET indicator were considered for the labor score but ultimately excluded. The scoring direction of NEET is ambiguous in the Turkish context: a high rate could mean either an untapped workforce or a structurally underqualified labor pool. Furthermore, since skilled IT personnel are mobile and labor is intentionally given a low weight, adding a weakly-signaling indicator would introduce noise without improving discrimination between provinces.

Nuclear energy. Nuclear capacity was considered as a third component of the renewable energy score. However, Türkiye has only one partially operational nuclear site (Akkuyu in Mersin, 4 800 MW planned), which would reduce the criterion

to a near-binary signal rewarding a single province. We adopt the principle of including a criterion only when a complete, province-level dataset is available.

Wind data recency. The wind component uses the TÜREB July 2022 report, as more recent full editions are restricted to members. Provincial rankings of installed capacity have remained stable, but absolute values may be conservative for fast-growing provinces. This is a straightforward future update.

Deterministic model. All parameters are treated as fixed. In reality, energy prices fluctuate, seismic hazard estimates carry uncertainty, and population demographics shift over time. Monte Carlo simulation over uncertain parameters—re-running the solver thousands of times with sampled inputs—would identify which provinces are robust selections across many scenarios. This is the most natural next step for the project.

Cost model simplicity. The cost estimate uses a linear formula based on provincial GDP and a uniform electricity tariff. Real data center costs involve land prices, tax incentives, fiber optic connectivity, water availability, and many other factors we did not model.

VII. CONCLUSION

We presented a multi-criteria Binary Integer Programming model for data center placement across Türkiye’s 81 provinces. The project covers the complete pipeline: data collection from eight government and international sources, criterion-specific normalization (logarithmic scaling for seismic PGA, wind capacity, and labor force; inverted min-max for cooling and seismic safety), BIP formulation with four constraint families, exact solving via PuLP/CBC’s branch-and-bound algorithm, and interactive visualization through a Streamlit dashboard.

The central finding is that Central Anatolia consistently emerges as a strong region for data center placement under most weight configurations, offering moderate scores across all criteria without catastrophic weaknesses. However, the model’s strength lies in demonstrating how different priorities lead to meaningfully different selections: seismic safety pulls the solution to central Türkiye, cooling efficiency pushes it east, and renewable energy draws it to the Aegean coast. This sensitivity to user preferences is the model’s most practically useful feature.

For future work, three extensions would strengthen the analysis: (1) Monte Carlo simulation over uncertain parameters to quantify robustness; (2) finer-grained spatial data (parcel-level rather than province-level) to improve accuracy; and (3) additional criteria such as internet backbone proximity and water availability to capture more dimensions of the siting decision.

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